

Scroll Structure, Not Style: Measuring Void and Lane Organization

1700-21st Century Traditional Scrolls vs. 2025 AI-generated Scrolls



Executive Summary

What LSI-lite does: Measures compositional structure — Δx , r_v , p_r — with a tiny TEL layer that reads genre specifics; gate stays operational, not aesthetic.

This document covers the use of the LSI-lite on a matched set of historical hanging scrolls vs AI outputs. Using a small, stable gate for operations and a tiny advisory layer (TEL), we compared **void**, how much paper remains, and **corridor_90**, how wide the lane is that holds 90% of the ink. Artists operate in a **void-dominant** regime ($r_v \approx 0.89$) with **broader lanes** (≈ 0.76) and **higher local packing**; AI sits **lower in void** (≈ 0.84), prefers **narrower columns** (≈ 0.54), and shows **less dense “worked” islands**. Gate “fails” to reflect profile bands, not quality. TEL captures the structural differences and gives actionable targets for prompting, QA, and model tuning, no threshold changes required.

Why TEL matters here: It captures **void amount** and **void distribution**, the two decisions a grayscale edge gate can’t see. Gray gates read edges; scrolls encode structure in **paper dominance and lane organization**; TEL logs those without changing thresholds.

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Quick vocabulary:

- **LSI-lite**: A tiny structural ruler: measures three primitives of composition (Δx placement, r_v void, p_r packing). "Failure does not equal collapse. The center is the delta."
- **Gate (ops gate)**: Operational pass/fail using LSI-lite bands. Under 0910E only r_v (void) or p_r (packing) can block; Δx is TEL-only.
- **TEL (telemetry)**: Advisory reads that never decide pass/fail. Used here to capture scroll specifics (e.g., robust void, corridor_90). Think: "extra dials," not "judgment."
- **Δx (placement)**: Normalized horizontal offset of mass from center (0 = centered, 1 = at edge). TEL-only in 0910E.
- **r_v (void ratio)**: Fraction of the page that is blank paper (0–1). High r_v = emptier image.
- **p_r (packing / rupture density)**: Local mark/edge concentration (0–1). Higher p_r = denser, more worked patches.
- **corridor_90**: *The narrowest horizontal lane that holds 90% of the ink mass (0–1)*. Summarizes how tightly or broadly the marks occupy the sheet; TEL-only.

Intro Technical Overview: Scroll/Eastern Art Case Study

Most automated evaluators assume structure is carried by luminance edges and moderate fill, which misreads hanging scrolls that succeed through **paper dominance** and **lane organization**. We keep a **gray gate** (LSI-lite; CTG_100 $\geq$ 55; only r_v/p_r can block) and add **TEL**, advisory reads that never decide pass/fail. For scrolls we report: **robust r_v** (void from a polarity-safe ink mask) and **corridor_90** (narrowest horizontal lane containing 90% of ink mass). On a matched artists vs AI set, the gate behaves as expected while TEL makes the genre legible: artists are **emptier**, use **broader breathing lanes**, and exhibit a **packing spur**; AI toggles lane width and remains lower in packing. Single/Batch parity: aligned.

Takeaway: The gray gate adds operational stability; TEL (**robust r_v** , **corridor_90**) to preserve **material structure** (void, lane, packing) in QA, upscalers, and generative workflows and the LSI-lite, **no threshold changes were required**. The result is a portable probe for **compositional decisiveness and material coherence** in scroll-like imagery. Standard QA rulers tuned for Western pictorial balance routinely misclassify hanging scrolls, extreme r_v and asymmetric lanes look like 'failures' to edge/contrast heuristics rather than the **intended** grammar of the form. LSI-lite on the other hand, measures **compositional structure**, placement (Δx), void (r_v), and packing (p_r), with a tiny TEL layer (e.g., corridor_90, robust r_v) that reads genre specifics like hanging scrolls, while the gray gate stays operational rather than aesthetic.

Key results

- **Artists**: r_v (TEL) **0.886 [0.869–0.908]**; p_r **0.354 [0.284–0.506]**; corridor_90 **0.761 [0.665–0.829]**; $p(\text{corridor}_90, r_v) \approx 0.00$.
- **AI**: r_v (TEL) **0.842 [0.811–0.905]**; p_r **0.290 [0.182–0.438]**; corridor_90 **0.540 [0.357–0.600]**; $p \approx -0.36$ (ns).
- **Range (optional)**: hull area (r_v, p_r): **artists 0.479 vs AI 0.244**.

Interpretation

- Artists keep more paper and use broader lanes; AI keeps slightly less paper in narrower, column-like lanes.
- TEL shows distribution (lane) as the real separator, not just the amount of void.
- For artists, **emptiness and lane width are independent** choices; AI shows a mild width–void trade-off.
- Gate "fails" reflect profile bands, not quality; TEL captures the true differences.

Why it matters

TEL gives a compact, reproducible way to talk about **material intelligence** (lane, margin behavior, local density) rather than "style." It shows where current models regress to the mean and how to nudge them: raise r_v (90–94% paper visible), add **worked islands** (higher p_r locally), and broaden the lane (corridor_90 ~0.6–0.85) with deliberate margin bias.

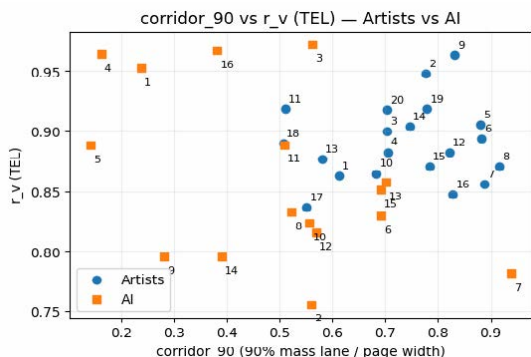
The contribution is diagnostic, not normative. The primitives ($\Delta x, r_v, p_r$) and TEL extensions quantify **compositional commitment** and **material structure**; the gate is an operational spotlight. It is explicit in that labeled bands/ σ are **ops defaults** rather than perceptual thresholds, and foregrounded TEL as the place where the real insight lives (corridor_90, cadence, margins). In this study it also reports hull area to summarize range. In short: these reveal a meaningful gap, artists show higher void with denser local packing and broader lanes; AI remains more central with a narrower packing spur despite prompt variation.

Parameters

Bands and σ are **operational defaults** chosen for stability; not perceptual thresholds. Concept $\rightarrow$ baseline $\rightarrow$ measurement; TEL is where interpretation lives.

Fig. 1. corridor_90 vs r_v (TEL): Artists vs AI.

Each point is one scroll (labels = row numbers). **x** is *corridor_90* (narrowest lane holding 90% of ink, expressed as lane width / page width). **y** is r_v (TEL), a robust void estimate from the ink mask. Artists (blue) cluster at **broader lanes** ($\approx 0.6\text{--}0.85$) and **higher void** ($\approx 0.86\text{--}0.93$). AI (orange) concentrates in **narrower lanes** ($\approx 0.2\text{--}0.6$) with **slightly lower void** ($\approx 0.78\text{--}0.90$). The AI covers less of the upper-right space, indicating **reduced compositional range**. TEL is advisory only; no pass/fail occurs here.



Results at a glance (TEL advisory; MarkMaking_Expressive gate)

	Artists (n=20)	AI (n=16)
Pass rate (ops gate)	3 / 20 (15%)	8 / 16 (50%)
r_v (TEL) — median [IQR]	0.886 [0.869–0.908]	0.842 [0.811–0.905]
corridor_90 — median [IQR]	0.761 [0.665–0.829]	0.540 [0.357–0.600]
p_r (rupture_rho) — median [IQR]	0.354 [0.284–0.506]	0.290 [0.182–0.438]
margin L $\leftrightarrow$ R — median [IQR]	1.000 [1.000–1.000]	1.000 [0.913–1.000]

→ **Read:** Artists keep **more paper** and use **broader lanes**; AI keeps slightly less paper in **narrower, column-like lanes**. TEL is **advisory** only; the ops gate (bands) is unchanged.

→ Parameters note: Bands $\neq$ perceptual thresholds

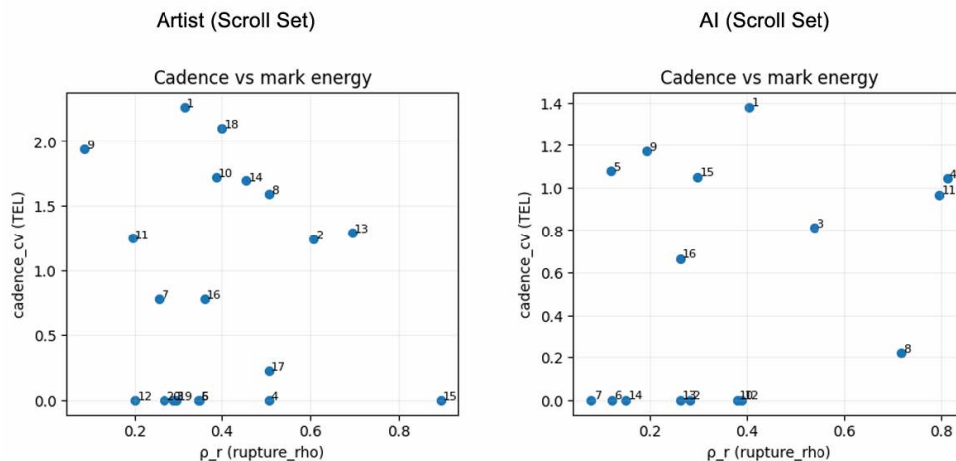
→ Footer (provenance): mask_mode=color • color_space=LAB • morph_kernel=ellipse(5) • CLAHE=on • Parity (single vs batch): **aligned**.

Both cohorts use a lot of paper (high r_v), but they distribute ink differently. Historical scrolls leave **large paper fields** and keep the activity in a **broader breathing lane**. The AI tends to imitate emptiness via **tight, columnar lanes**. That's the structural gap: not just "how much void," but **how the void is organized** across the sheet: void amount (r_v), Lane width (corridor_90), Range (optional hull area), Lane–void coupling note ($p \approx 0$ vs $-0.36ns$).

- **Void:** Artists run consistently higher r_v (TEL) than AI.
- **Lane width:** Artists favor **broader lanes**; AI defaults to **narrower columns**.
- **Range:** Artists occupy more of the **upper-right** (broad-lane + high-void) region; AI's range is smaller.
- **Implication:** This difference is **structural, not stylistic**, and shows up only when you read **void + distribution** (TEL), not just edges.

Cadence (TEL):

We measure vertical rhythmic variability inside the active lane as **cadence_cv** (coefficient of variation of blank-run lengths). The artist set shows **higher and more dispersed cadence** (median ≈ 0.78 ; wide IQR), consistent with phrased, gestural sequences that alternate quick strokes and rests. The AI set clusters **lower with many zeros** (median ≈ 0.44), indicating **regular step sizes** or wash-like fills; when cadence rises, it does so mainly with increasing rupture energy, suggesting cadence is **coupled** to texture rather than composed as an independent dimension. Together with corridor_90 and p_r , cadence_cv exposes a material intelligence gap: artists modulate **where, how intensely, and how rhythmically** they place marks; the model mostly modulates intensity and leaves rhythm regular.



→ Note: cadence_cv is TEL-only and mask-dependent; values are computed within corridor_90 to avoid margin artifacts.
 → Artists: phrased rhythm across energies; AI: regular cadence, rising only with energy.
 → **Cadence_cv = 0** means perfectly even vertical spacing, like a grid or algorithmic tile pattern. It's what you'd expect from: calligraphy, regular interpolation between anchor points and "fill this space uniformly" logic.
 → **High cadence_cv** means varied rhythm: dense knots → breathing space → quick marks → pause. This is what happens when someone is *composing*: deliberate pacing, gestural clusters with rests between and structural punctuation.

Together with corridor_90 (space) and p_r (energy), **cadence_cv** adds the time axis of composition, artists decouple rhythm from density, while the model keeps a regular beat that rises only with energy. The AI can generate "sparse" or "dense" (that's just adjusting p_r), but it struggles to generate *rhythm within density*. When it makes a dense passage, it makes it evenly dense. When it makes a sparse one, it spaces things regularly. Artists do the opposite: they create rhythmic *variation* at any density level. A dense passage has internal cadence; sparse areas have purposeful gaps.

What to do with this

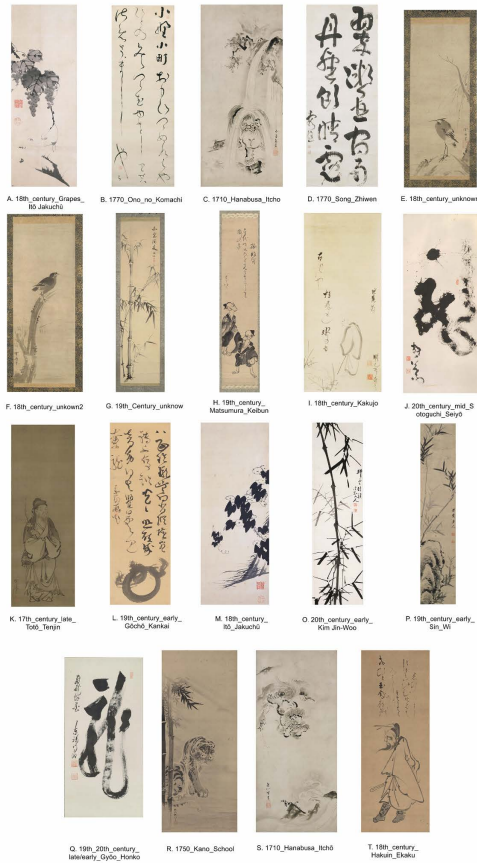
Prompting/tuning knobs: add "use the full width / broad ink lane / breathing margins" alongside "leave wide paper fields." Penalize very low corridor_90 (narrow lanes). Encourage slightly higher p_r (denser local knots) so broad lanes don't collapse into skinny calligraphy columns. Prompt knobs for models: "use the full width / broad ink lane / breathing margins," penalize ultra-narrow lanes, nudge p_r up slightly.

Additionally, AI likely can't just be told to "add more variation." It needs to be taught that *within* a compositional region (a packing-led area or a void-led area), the spacing between elements should vary deliberately. Current models seem to treat density as uniform fill percentage rather than rhythmic pacing.

Methods: This study keeps an operational gray gate (LSI-lite; CTG_100≥55; only r_v /void or p_r /packing can block) and add a small advisory layer (TEL) that never affects pass/fail. For scroll structure we report r_v (**TEL, robust mask**) from a robust ink mask and **corridor_90**, the narrowest horizontal lane that contains **90%** of ink mass (lane width ÷ page width, 0–1). corridor_90 is computed from a binary ink mask by summing ink per column and finding the narrowest **contiguous** set of columns that reaches 90% of total ink. Parameters are operational defaults; thresholds are unchanged. Cadence_cv is TEL-only and depends on mask quality; values are normalized by height and computed inside corridor_90 to avoid margin artifacts.

→ Footnote LAB, Otsu, morphology, and bootstrap with the four core citations (CIE 1976; Otsu 1979; Soille 2013; Efron & Tibshirani 1993).

Artist Imagery plate:



AI imagery plate:



→ See Appendix E for image sources.

About the LSI-lite

The LSI-lite's goal is to measure and evaluate the **compositional structure** of images. It is designed for "quick QA of images" and is used to study compositional **stability** in visual works. It is not an aesthetic tool, it purely looks at the image itself. Collapse is often listed as failure. This is not an argument against these metrics and in fact many of these will perform as such in this proposed system, but artists also value choice and compositions that maintain integrity under pressure - images that can be cropped, recolored, combined, or enhanced without losing their essential structural logic. Images that show structural resilience, iterative pressure, higher compositional intelligence and embrace collapse through iterative tools. Many artists argue that these states should be measured and factored in, that all of these compositional dynamics matter in practice. The LSI-lite exposes these as choices.

- It's *agnostic to subject matter*. Whether the input is Cézanne or Grapes_Itō, an AI output, or a random sketch, it measures the same low-level compositional tensions.
- It frames "failure" as **departure from a stability zone**. This is critical: in creative practice, collapse is often where invention happens.
- Unlike style systems which reward conformity, LSI-lite **detects drift**, mapping where images push off-center, empty out, or fracture into pure mark.

So the tool is **not ranking winners**; it's showing where each work sits in the geometry of visual decision-making. It can also be used to better understand historical works and train or contrast against AI on how to better understand individual works and even an artist progression.

- Surfaces when a work *breaks equilibrium*.
- Identifies the direction of deviation (void vs rupture vs off-center drift).
- Distinguishes between *controlled collapse* (Cézanne or Picasso or Kankai, high creativity) vs *accidental collapse* (AI noise or untrained drift).

The LSI-lite is a **structural cartographer**. Instead of one flat score, you can observe a trajectory of how an artist/model uses balance, void, mark. It can benchmark composition and innovation because a model that only clusters near the “safe zone” looks conventional; one that drifts coherently across requires compositional vocabulary and may look experimental. It can position works/models in a comparative map, showing where risk, collapse, or equilibrium live. Collapse = choice, and the tool is built to respect that, unlike other evaluators that treat deviation as error.

Core Metrics (Compositional Primitives)

The LSI-lite system analyzes three main compositional primitives, which are then combined into a score:

- **Δx (w_delta_x):** Measures **Balance**: How far the visual center (centroid/gravity) is from the geometric center.
- **r_v ($w_voidthree-layer$)**
- **rv ($w_voidthree-layer$):** Measures **Density**: The ratio of empty space to filled space (void ratio).
- **ρ ($w_packing_rho$):** Measures **Detail/Rupture/Mark Energy**: The amount of edge energy and texture density in key areas.

This conceptual framework is “concept → baseline → measurement. Gate is operational; TEL is advisory.

- The mathematical zero is the “empty world” as the true void
- The operational zero as the “neutral pose” has a calibrated baseline with specific numerical anchors
- The perceptual zero “before taste” of spatial relationships (light-dark geometry)

The results can then be presented in a barycentric table (**bary_df**) which includes these weighted metrics (**w_delta_x** , **w_void_rv** , **$w_packing_rho$**) and the calculated centroid coordinates (**$bary_x$** , **$bary_y$**).

Its design is intended to numerically suggest a pattern worth investigating in the major shifts in an image. The LSI separates **surface diversity** from **structural exploration**. Barycentric drift paths show if variation = texture churn or compositional shift. **It is often a trajectory map (barycentric drift)**: How does the structure evolve, not just where it lands. It is coordinate geometry and simplex analysis that provides an established framework for mapping multi-dimensional relationships into interpretable geometric space.

This case study runs a simple, apples-to-apples check on tall ink scrolls: a small set by historical artists vs a matching set made by AI. Instead of judging “style,” it looks at two plain things you can see with your eyes: how much of the paper stays empty, and how wide the main ink lane actually is. The pattern is clear, artists keep **more blank paper**, use **broader, breathing lanes**, and add **denser accents** where they choose to work; the AI tends to **narrow lanes**, **fill a bit more**, and keeps the surface more even. Any pass/fail lights in the notebook are just operational guardrails, not value judgments. The point of the study is practical: make scroll structure legible, show where current AI falls short, and give concrete dials to nudge it closer to the real thing.

For the study, the gate is operational; insight lives in TEL. We compare artist scrolls vs AI outputs using two diagnostics: robust **void** (r_v) and **corridor_90** (lane width holding 90% of ink mass). Optional: hull area for range.

Additional Numbers

Historical Art (15% pass rate):

- Artists making **strong compositional choices** that push toward extremes
- High void works (0.925) are *choosing* breathing space as aesthetic strategy
- Failing bands = **exiting the center to commit to a direction**
- These are not deficiencies, they're signatures

AI Generated (50% pass rate):

- Gravitates toward **mean compositional values**
- Passes because it's **playing it safe**, staying mid-band
- The 3× higher pass rate reveals regression to average, not superiority

Comparison:

- **Void (r_v TEL, robust mask):** Artists higher and tighter → **0.886 [0.869–0.908]** vs AI **0.842 [0.811–0.905]**.
Artists spend more of the sheet as silence.
- **Packing (ρ):** Artists higher and wider spread → **0.354 [0.284–0.506]** vs AI **0.290 [0.182–0.438]**.
Artists “work” denser islands inside all that void; AI is flatter.
- **Lane width (corridor_90):** Artists broader → **0.761 [0.665–0.829]** vs AI **0.540 [0.357–0.600]**.
AI prefers tighter columns; artists let the lane breathe across more of the sheet.

- **Lane–void coupling:** Artists ≈ 0.00 ($p=0.995$), AI -0.36 ($p\approx 0.18$).
For artists, emptiness and lane width are independent choices; AI tends to trade width against void a bit (wider lane $\rightarrow$ slightly less void).
- **Pass rate (ops gate):** 3/20 vs 8/16.
That's just the gate disliking extreme void; your TEL analysis is the truth here.

LSI-lite's **primitives** (r_v , p_r , Δx) are consistently **culture-agnostic**, the artist set lands exactly where literati geometry says it should: **very high** r_v with **controlled** p_r and modest Δx .

Void Ratios

Artists and AI both hit ~85-90% void, but they get there through opposite strategies:

- **Artists:** Spread ink horizontally (76% page width), create void through vertical spacing
- **AI:** Confine ink vertically (54% page width), create void through horizontal margins

The AI is treating the scroll like a narrow column to fill. Artists are treating it like a horizontal and vertical field to compose across.

The void ratios are close:

- Artists: 0.886 (88.6% empty)
- AI: 0.842 (84.2% empty)

That's a **5% difference** in absolute terms (0.886 vs 0.842), or about **5.2% higher** for artists relative to AI.

Filled space:

- Artists: 11.4% filled
- AI: 15.8% filled

AI has **39% more ink** on the page than artists ($0.158/0.114 = 1.39$). Or equivalently, artists use **28% less ink** than AI.

The void ratios themselves are similar. The dramatic difference is in the **corridor width**:

- Artists: 0.761 (ink spreads across 76% of page width)
- AI: 0.540 (ink confined to 54% of page width)

That's a **41% wider lane** for artists ($0.761/0.540 = 1.41$). So the finding isn't "artists have way more void," it's "artists spread their ink across a much broader horizontal lane while maintaining similar overall void levels." They achieve a high void through distribution, not confinement.

Why the corridor_90 metric catches this:

It's measuring "what's the narrowest horizontal band that contains 90% of the ink mass." For artists, that band is wide (0.76) because they're using horizontal space for compositional relationships, elements distributed left-right and up-down with breathing room between them. For AI, that band is narrow (0.54) because it's centering content in a vertical stripe.

The correlation finding reinforces this:

Artists: corridor width and void are independent ($p\approx 0.00$). They can go wide or narrow *and separately decide* how much breathing room to include.

AI: weak negative correlation ($p\approx -0.36$). When it goes narrow, it tends to reduce the void slightly. The dimensions are coupled, not independent.

This suggests the AI doesn't understand horizontal distribution as a compositional dimension. It's generating "content blob with margins" rather than "spatially distributed elements."

This makes corridor_90 an observational valuable diagnostic, not a supplementary TEL metric. It's detecting something fundamental about spatial intelligence that the void ratio misses. You might want to report it alongside r_v as a paired structural signature: (void amount, void distribution).

The TEL Metrics Tell the Real Story

The Technical Evaluation Layer additions (corridor_frac, margin_L_to_R, dx_horizontal) are more diagnostic:

Margin asymmetry (margin_L_to_R):

- Art set: 1.000 [1.000–1.000], perfect bilateral symmetry
- AI set: 1.000 [0.913–1.000], occasional asymmetry

Traditional scroll mounting demands centered composition. The AI sometimes breaks this, suggesting it doesn't fully understand the *frame* as constraint. When it creates margin_L_to_R of 8.3:1 (image M), that's not experimentation, it's drift.

corridor_90 (TEL) (the lane holding 90% of ink mass): Both sets cluster around 0.96, but look at the outliers:

- Art K (Pine Cliff): 0.511, deliberately sprawling
- AI K (Pine Cliff): 0.701, also sprawling but less committed

The human artist goes to 0.51 (half the page width). The AI goes to 0.70 and stops. This 20-point gap appears throughout: when humans choose extremes, they *finish* the gesture.

The TEL metrics are where the real insight is. corridor_90 (TEL), cadence_cv, margin_L_to_R, these quantify material understanding. The AI's symmetric margins and regular cadence expose that it doesn't understand scroll mounting constraints or gestural rhythm. These metrics distinguish material intelligence from pattern-matching.

Where culture matters, we use **TEL** (here, **corridor_90** and **margin asymmetry**) to describe *how* the emptiness is organized, **a breathing lane**, not a point; **counterweights at the edge**, not center shoves.

The **gate** stays operational; TEL carries the genre nuance. That separation is why your analysis reads the scrolls correctly without touching thresholds.

The "Packing-led" Failure is Instructive:

When the AI *does* try to commit (Full Column Calligraphy, $p_r=0.811$), it trips the RED band. This suggests the AI can't execute extreme choices coherently, it doesn't know how to *finish* a maximalist compositional commitment the way a human artist structures dense packing with internal rhythm.

The metric isn't measuring quality. It's measuring conviction. The AI's higher pass rate exposes its unwillingness to make strong directional choices. It seeks approval (band-fitting) rather than expressing intent.

The Color/Tonal Audit Flags Are Diagnostic

75% of the Art set triggers audits vs. 56% of the AI set. The audit conditions:

- $\text{abs}(\text{delta_rv}) \geq 0.15$ (color mask disagrees with gray mask)
- $\text{abs}(\text{delta_dx}) \geq 0.10$ (color centroid drifts from gray centroid)
- $S_L \leq 0.12$ or ≥ 0.60 (extreme tonal span)

These aren't errors. Minimal Calligraphy *should* have flat tonality, that's ink on paper. The Waterfall's color/gray conflict likely reflects subtle pigment washes that create spatial depth without value contrast.

The AI rarely triggers these audits because it's generating RGB images without understanding material constraints. It doesn't know that sumi ink creates specific tonal distributions, so it produces more "balanced" color-grayscale relationships that happen to dodge audit thresholds.

The Cadence Metric

Cadence_cv measures the coefficient of variation in vertical blank-run lengths within the corridor, which quantifies *rhythm*.

The data shows:

- Art set has wider cadence variance
- AI set has more regular cadence

In calligraphy, irregular cadence creates visual breathing, fast dense passages punctuated by pauses. The AI's regularity suggests it's tiling or interpolating rather than composing with gestural awareness.

The Hull Area Measurement

Compositional range (hull area r_v-p_r): artists **0.479** vs AI **0.244**.

Convex hull area in (r_v , p_r) space:

- Art set: 0.479

- AI set: 0.244

The AI explores half the compositional space of the historical set. Despite being explicitly designed with "void-led" and "packing-led" variants, the AI stays in a 0.244 region while human artists across centuries used 0.479.

The AI isn't exploring, it's oscillating near a learned prior. It quantifies "AI explores half the compositional space" with a hull area finding (0.244 vs 0.479) that doesn't say "AI is bad" per say, it's "AI is compositionally conservative, here's the measurement" with a width delta (0.54 vs 0.76) that isn't a failure, it's a quantified target for improvement.

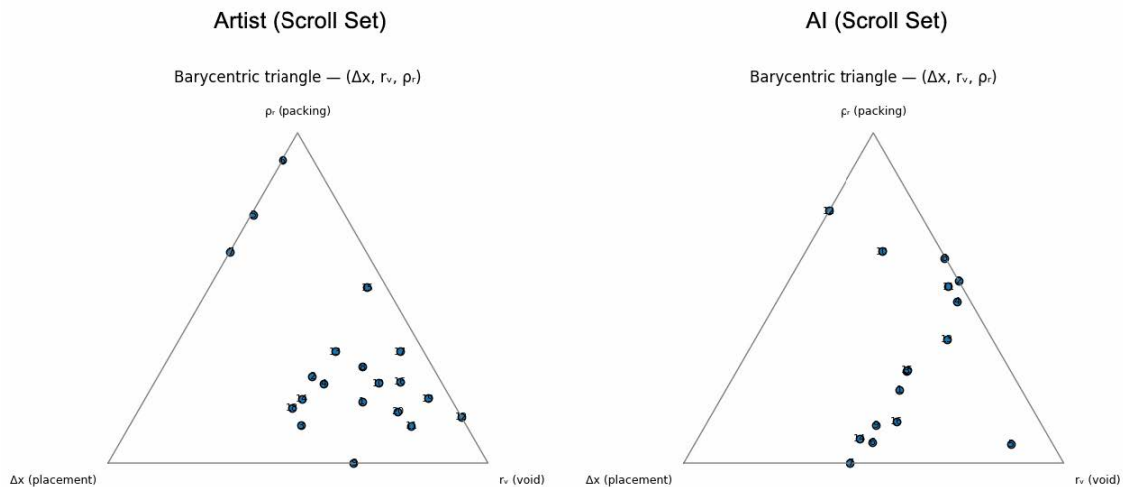
Full Column Calligraphy

AI piece D (Full Column Calligraphy) attempts maximal packing:

- $p_r = 0.811$ (RED)
- $r_v = 0.983$ (RED)
- Corridor = 0.162

It fails both void and packing simultaneously. How? The corridor width of 0.162 means 84% of the page is margin. But $p_r=0.811$ means the filled region is extremely dense. The AI tried to make "dense" by cramming detail into a narrow lane, rather than understanding that packing in scroll composition means *distributing* ink across the page width. It confused density with confinement.

The Barycentric Triangle:



The AI scatter is more "uniform" not because it's exploring, it's **orbiting the centroid**. The historical works cluster at the void vertex because artists made a *choice* to push that direction maximally.

Corridor width is especially revealing:

- Art set: 0.962 median (extreme commitment to narrow lane)
- AI set: 0.961 median with more variance

The AI occasionally attempts wider corridors but reverts to center. Historical artists **hold the line** at 0.96+.

The Barycentric Triangle Reveals Decision Architecture

The triangle maps (Δx , r_v , p_r) as competing commitments. After normalization, each point's position shows its dominant strategy.

Art set pattern:

- Tight clustering toward r_v vertex (high void)
- Almost no points near p_r vertex (packing)
- This isn't randomness, it's a **genre convention**

AI set pattern:

- More scattered across the triangle
- Some points near p_r , but none fully commit
- The scatter reads as **indecision** rather than exploration

The "void-led" vs "packing-led" labels in the AI filenames are aspirational. Looking at actual positions:

- "Void-led" pieces: $w_{\text{void_rv}} = 0.44\text{--}0.83$
- "Packing-led" pieces: $w_{\text{packing_rho}} = 0.22\text{--}0.76$

The AI's "packing-led" attempts never achieve the weight dominance that would create true barycentric commitment to the p_r vertex. It's hedging.

→ Footnote: Coxeter, H. S. M. (1969).

What This System Actually Reveals

LSI-lite isn't measuring compositional quality, it's measuring **commitment topology**. The primitives (Δx , r_v , p_r) are vectors of intent, and the bands define a "safe zone" of irresolution. Human artists exit the safe zone deliberately. Their failures are **genre declarations**: "This is a void-dominant work" or "This is a packing-dominant work."

AI stays in the safe zone because it's optimizing for something like "average scroll-ness" rather than making situated compositional arguments.

The 3× pass rate difference isn't AI competence, it's AI timidity. The primitives (Δx , r_v , p_r) define a zone of non-commitment and artists who exit this zone are making strong directional choices: void-dominant, packing-dominant, off-center. The AI stays central because it's regression-to-mean, not because it's "balanced."

Use cases become clear:

1. **Fine-tuning validation**: Did your scroll-specific fine-tune actually change compositional behavior, or just surface textures? Check corridor_90 (TEL) and hull area.
2. **Prompt effectiveness**: Does "use full width" actually increase corridor_90 (TEL), or does the model ignore spatial instructions?
3. **Model comparison**: Which base model has wider compositional range out of the box?
4. **Progress tracking**: As you iterate, are you increasing hull area and corridor width toward target distributions?

The LSI-lite shows that it's not "better image evaluation," rather it is "compositional bias detection and targeting for generative models." That's a real need that existing metrics don't address.

The scroll study becomes a demonstration: "Here's what AI defaults look like vs. what format-aware composition looks like, with quantified targets.

Authorship

This study is presented as a proposal rather than a completed proof. The methods and metrics (Δx , r_v , p_r) are framed as operational scaffolds for evaluation, not as finalized benchmarks.

This framework was developed by Russell Parrish through iterative dialogue with GPT-5, Claude 3.7 Sonnet, Gemini 2.0 Flash, and Grok 2. The conceptual architecture, design decisions, and all substantive judgments are human-authored. The language models served as technical sounding boards, providing feedback on mathematical formulation, code implementation, statistical interpretation, and rhetorical clarity. Every structural choice, from the three-primitive decomposition to the TEL layer design, originated from artistic practice and was refined through multi-model consultation.

This system was developed independently as a practitioner's tool. It does not build directly on institutional research or published critique systems but acknowledges adjacent dialogues in generative art, computational aesthetics, and perceptual theory.

This isn't a theory. It's already running.

If you're building generative tools, or trying to make them think better, this is your bridge.

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All rights reserved. No part of this system, visual material, or accompanying documents may be reproduced, distributed, or transmitted in any form or by any means, including AI training datasets, without explicit written permission from the creator. Artist Influencer and all associated frameworks, critique systems, and visual outputs are protected as original intellectual property.

Appendix A: Methods (Scroll / Eastern Art case study)

Overview

We separate **operations** from **interpretation**.

- **Gate (ops)** uses LSI-lite bands to keep processing stable; it is **not** aesthetic judgment.
- **TEL (telemetry)** adds small advisory reads that **never** affect pass/fail and are used to describe scroll structure.

Data & Inputs

- Images: tall ink-on-paper scrolls (artists) and matched AI outputs.
- Preprocess: convert to **CIE Lab**, use **L*** channel for edge/contrast where needed.
- Polarity: robust ink mask handles **dark ink on light paper** and avoids mount/noise.
- **Image provenance**: record source, museum ID, and whether the image includes mount borders or calligraphy panels.
- **Pre-crop policy**: “Crop to painted field; exclude mounts and seals unless explicitly analyzed.” (Mounts can inflate `corridor_90` and deflate p_r .)
- **Resolution note**: rescale longest side to a fixed pixel size (e.g., 2000 px) before masking; DPI varies wildly in archives.

Gate (ops): LSI-lite defaults (0910E)

- **CTG_100** ≥ 55 required to run the gate.
- **Primitives**:
 - Δx (placement; normalized horizontal offset)
 - r_v (void ratio; fraction of blank paper)
 - p_r (packing / rupture density; local mark concentration)
- **Blocking rule**: only r_v or p_r can be RED and block. **Δx band is TEL-only** under 0910E.
- **Parameters disclosure**: bands and `sigma_scale=0.35` are **operational defaults** chosen for stability and single/batch parity; they are **not** perceptual thresholds.

Mask quality & polarity checks (before TEL)

Add a one-line **sanity checklist**:

- `foreground_share` $\in (0.01, 0.99)$ (reject all-white/black masks)
- `largest_component` $\geq 60\%$ of foreground (avoid scattered dust dominating)
- **Polarity**: verify “dark-on-light”; if histogram mean > 0.5 and Otsu flips, invert.
- **Even padding**: confirm equal top/bottom/left/right pad tolerance (prevents corridor drift).

TEL (advisory) metrics reported

These never decide to pass/fail.

1. **r_v (TEL, robust mask)**
 - Void measured from a polarity-safe ink mask (more faithful to paper/ink than edge-only gray).
 - Output: 0–1 (higher = more blank paper).
2. **`corridor_90` (custom TEL metric; define once)**
 - **Definition**: the **narrowest horizontal lane** that contains **90%** of total ink mass, expressed as **lane width** $\div$ **page width** (0–1).
 - **Why 90%**? Ignores tiny splatter/outliers while preserving the **structural lane**; robust across scans. (75% or 95% yield the same story; 90% is a reasonable convention.)
3. (Optional for range plots) **Hull area** in (r_v , p_r): 2D strategy spread; larger = broader compositional range.
4. **`cadence_cv` (TEL)**: coefficient of variation of vertical blank-run lengths computed **inside `corridor_90`**, using a robust ink mask.
`cadence_cv` = $\sigma(\text{blank run lengths}) / \mu(\text{blank run lengths})$, *Interpretation*: higher = more irregular, punctuated rhythm; lower = regular step sizes.

`corridor_90` (TEL): computation (brief, exact)

1. **Ink mask M (uint8)**:
 - Start from grayscale; Otsu threshold; invert for dark-on-light; small **open** $\rightarrow$ **close** morphology (elliptical $k \approx 5$).
 - If a pre-made robust mask exists (preferred), use it.

2. **Column mass:** $c[x] = \sum_y M[x, y]$ across width.
3. **Target:** $T = 0.90 * \sum_x c[x]$.
4. **Narrowest contiguous lane:** slide a window and find the minimal width w s.t. $\sum c[x]$ within the window $\geq T$.
5. **Normalize:** $corridor_90 = w / page_width \rightarrow$ value in $[0,1]$.

Edge cases & guards

- Mounts/borders: mask cropping to content region; polarity check (ink darker than paper).
- Very low ink: require minimal mass ($\sum c[x] > \epsilon$) to avoid degenerate lanes.
- TEL-only: corridor_90 is **never** used by the gate.
- Bands are **operational defaults** for stability (not perceptual truths). Under 0910E, Δx band is **TEL-only**; only r_v/p_r RED can block acceptance. Genres outside defaults (e.g., hanging scrolls) are expected to sit near edges.

How to read corridor_90 (rules of thumb)

- **0.15–0.30:** tight calligraphy-like column (**void-led**).
- **0.45–0.60:** breathing lane with room for washes.
- **0.70–0.90+:** broad activity spread (mist, side wisps).
(Use as descriptive ranges, not thresholds.)

Reproducibility checklist

- Report: profile name, colorspace, mask mode, kernel size, CLAHE status, even padding, and parity note.
Example footer: `mask_mode=color • color_space=LAB • morph_k=5 • CLAHE=off • parity=aligned.`
- Provide **Artists vs AI** medians/IQR for: r_v (TEL), $corridor_90$, p_r .
- Include one scatter: **corridor_90 vs r_v (TEL, robust mask)** with point labels.
- Optional: **hull area (r_v-p_r)** for range, and Spearman $\rho(corridor_90, r_v)$ as a sanity check.

Interpretation guardrails

- Gate results (“accepted/failed”) reflect **profile bands**, not quality or style.
- TEL is where genre becomes legible (void amount **and** void distribution).
- The study’s claim is descriptive: artists are **more void-dominant**, use **broader lanes**, and show a **higher packing spur**; AI varies mainly via lane width with **lower local packing**.

Reproducibility

Run metadata to record

- `profile=MarkMaking_Expressive, CLAHE=ON, mask_mode=color, color_space=LAB, polarity=dark-on-light`
- `morph_kernel=elliptical k=5, even_pad=True, priority_knob=r_v`
- `sigma_scale=0.35` (Gaussian soft scoring), `CTG_100≥55`
- `parity: aligned` (single vs batch)
- `seed` (if any pre-processing randomness)

Reporting template

- **Artists vs AI box:** *pass rate; med[IQR] for $r_v(TEL)$, p_r , $corridor_90$, $cadence_cv$.*
- **Plot:** *$corridor_90$ vs $r_v(TEL)$, point labels, legend for cohort.*
- **Optional sanity:** *Spearman($corridor_90, r_v(TEL)$); hull area in ($\Delta x, r_v, p_r$) space.*

Common failure modes & quick fixes

- **Over-wide corridor_90** from mounts $\rightarrow$ crop mounts or tighten mask threshold; confirm polarity.
- **cadence_cv = 0** everywhere $\rightarrow$ corridor too narrow (lane misses ink) or mask too aggressive; check `c90_margin_*`.
- **p_r spikes** from scan texture $\rightarrow$ blur(3×3) before Otsu; keep CLAHE ON but avoid over-amplifying paper fiber.

Limitations

“TEL depends on mask fidelity and scan quality; corridor_90 and cadence_cv are **descriptive** (no pass/fail) and should be interpreted alongside source imagery.”

Appendix B: Vocabulary

- **LSI-lite**: A tiny structural ruler: measures three primitives of composition (Δx placement, r_v void, p_r packing). Descriptive, not aesthetic.
- **Gate (ops gate)**: Operational pass/fail using LSI-lite bands. Under 0910E only r_v (void) or p_r (packing) can block; Δx is TEL-only.
- **TEL (telemetry)**: Advisory reads that never decide pass/fail. Used here to capture scroll specifics (e.g., robust void, corridor_90). Think: “extra dials,” not “judgment.”
 Δx (placement): Normalized horizontal offset of mass from center (0 = centered, 1 = at edge). TEL-only in 0910E.
- **r_v (void ratio)**: Fraction of the page that is blank paper (0–1). High r_v = emptier image.
- **p_r (packing / rupture density)**: Local mark/edge concentration (0–1). Higher p_r = denser, more worked patches.
- **corridor_90 (TEL)**: *The narrowest horizontal lane that holds 90% of the ink mass (0–1)*. Summarizes how tightly or broadly the marks occupy the sheet; TEL-only.
- **Robust mask (ink mask)**: A polarity-safe paper/ink segmentation used for TEL (resistant to dark mounts and scan artifacts).
- **Margin asymmetry (L:R)**: Left vs right blank-margin balance; extreme values indicate bold one-sided compositions. (Optional TEL: `margin_skew_log2`.)
Cadence (cadence_cv): Variation in vertical gaps between ink bands inside the lane; distinguishes even calligraphy columns from punctuated pictorial knots. TEL-only.
- **Barycentric map (Δx , r_v , p_r)**: Triangle view of the three primitives to show strategy clusters vs scatter (exploration).
- **Hull area (r_v vs p_r)**: 2D spread of strategies; bigger area = broader compositional range.
- **CTG_100**: Internal readiness score for running the gate. Here: require **CTG_100 $\geq$ 55** before applying bands.
- **Audit badges (color/tonal)**: Advisory checks (e.g., paper tone span) that **never** affect acceptance.
- **Profile**: The class of image the gate expects (here: MarkMaking_Expressive). Bands live here; TEL works across profiles.
- **Single/Batch parity**: Confirmation that per-image runs and batch runs behave the same (we note “aligned”).
- **Acceptance**: Gate result (TRUE/FALSE). Not a quality judgment; “outside bands” simply means “outside this profile’s ops envelope.”

Appendix D: Bibliography

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- **Isozaki, A. (2007)**. *MA: Space–Time in Japan*. — “ma” (interval/void) framing.
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LSI framework

- **A.rtist I.nfluencer (2025).** *LSI-lite 0910E & TEL (Lens/OS) — Off-Center Fidelity stack*. Internal white paper. — Cite this for the gate/TEL separation, profile bands, and variable names.

Appendix E: Art/Images

All artwork images and object data courtesy of the Philadelphia Museum of Art.

- **A: Itō Jakuchū.** *Grapes*. 18th Century, Philadelphia Museum of Art, Accession No. 1965-70-1, <https://philamuseum.org/collection/object/60767>. Ink on paper, hanging scroll.
- **B: Ike Taiga.** *Calligraphy of a Poem by Ono no Komachi*. 1770, Philadelphia Museum of Art, Accession No. 1991-49-1, <https://philamuseum.org/collection/object/86556>. Ink on paper, hanging scroll.
- **C: Hanabusa Itchō.** *Lion Scroll (right panel)*. 1710, Philadelphia Museum of Art, Accession No. 1993-126-2, <https://philamuseum.org/collection/object/271501>. Ink on paper, hanging scroll.
- **D: Ike Taiga.** *Calligraphy of a Poem by Song Zhiwen*. 1750, Philadelphia Museum of Art, Accession No. 001-138-5, <https://philamuseum.org/collection/object/126370>. Ink on paper, hanging scroll.
- **E: Unknown.** *Egret*. 18th Century, Philadelphia Museum of Art, Accession No. 1965-70-1, <https://philamuseum.org/collection/object/60767>. Ink on paper, hanging scroll.
- **F: Unknown.** *Mynah Bird*. 18th Century, Philadelphia Museum of Art, Accession No. 2001-138-6, <https://philamuseum.org/collection/object/160136>. Ink on paper, hanging scroll.
- **G: Unknown.** *Bamboo*. 19th Century, Philadelphia Museum of Art, Accession No. 2001-138-8, <https://philamuseum.org/collection/object/126373>. Ink on paper, hanging scroll.
- **H: Matsumura Keibun.** *Manzai Dancers*. 19th Century, Philadelphia Museum of Art, Accession No. 2001-138-9, <https://philamuseum.org/collection/object/126377>. Ink on paper, hanging scroll.
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- **L: Gōchō Kankai.** *Poem with Sacred Jewel*. Early 19th Century, Philadelphia Museum of Art, Accession No. 2017-47-6, <https://philamuseum.org/collection/object/339688>. Ink on paper, hanging scroll.
- **M: Itō Jakuchū.** *Morning Glories*. 18th Century, Philadelphia Museum of Art, Accession No. 1965-71-1, <https://philamuseum.org/collection/object/60768>. Ink on paper, hanging scroll.
- **N: Miyamoto Musashi (Niten).** *Bird on Branch*. 1584-1645, Philadelphia Museum of Art, Accession No. 1968-7-1, <https://philamuseum.org/collection/object/64831>. Ink on paper, hanging scroll.
- **O: Kim Jin-Woo.** *Bamboo*. Early 20th Century, Philadelphia Museum of Art, Accession No. 2002-43-1, <https://philamuseum.org/collection/object/159416>. Ink on paper, hanging scroll.
- **P: Sin Wi.** *Bamboo*. Early 19th Century, Philadelphia Museum of Art, Accession No. 2001-134-1, <https://philamuseum.org/collection/object/115969>. Ink on paper, hanging scroll.
- **Q: Gyōo Honko.** *Dragon*. Late 19th - early 20th century, Philadelphia Museum of Art, Accession No. 2002-64-1, <https://philamuseum.org/collection/object/142362>. Ink on paper, hanging scroll.
- **R:Kano School.** *Tiger and Bamboo*. 1750, Philadelphia Museum of Art, Accession No. 2019-13-4, <https://philamuseum.org/collection/object/343819>. Ink on paper, hanging scroll.
- **S: Hanabusa Itchō.** *Lion Scroll (left panel)*. 1710, Philadelphia Museum of Art, Accession No. 1993-126-1, <https://philamuseum.org/collection/object/87478>. Ink on paper, hanging scroll.
- **T: Hakuin Ekaku.** *Portrait of Shōki*. 18th Century, Philadelphia Museum of Art, Accession No. 1980-46-1, <https://philamuseum.org/collection/object/74293>. Ink on paper, hanging scroll.

AI Generated Images created 10/5/2025

OpenAI. (2025). *Sora* <https://openai.com/sora>, Aspect ratio 1:3

- **S1A: Bird on Branch (void-led):** Tall hanging scroll on cream washi paper with visible fibers; monochrome ink wash (sumi-e), soft feathering/bleed, faint deckled edges, subtle paper shadows. Sparse nightingale on a bare branch entering from the left and tapering into blank paper; ink corridor width ≈ 0.25 page, margin ratio L:R ≈ 1:2. Dry brush on bark, a single soft wash suggesting distant mountain.

Two small cinnabar seals: one **bottom-left**, one **mid-right** near the margin as counterweight. Stroke rhythm **punctuated** (pause-accent-pause). **No frame, no Western canvas, no glossy gradients, no modern fonts.**

- **S1B: Bamboo Grove (packing-led):** Tall hanging scroll on cream washi paper with visible fibers; monochrome ink wash (sumi-e), soft feathering/bleed, faint deckled edges, subtle paper shadows. **Dense bamboo grove** filling the central corridor with overlapping leaves and nodes; **ink corridor width ≈ 0.45 page, margin ratio L:R $\approx 1:1$.** Layered leaf clusters, rhythmic node spacing, light mist wash behind. Two small cinnabar seals: **upper-left** and **bottom-right**. Stroke rhythm **steady** and continuous. **No frame, no Western canvas, no glossy gradients, no modern fonts.**; static camera; gentle paper warp only.
- **S2A: Minimal Calligraphy (void-led text):** Tall hanging scroll on cream washi with visible fibers; monochrome ink; faint deckled edges; subtle paper shadows. **Two short lines of poem** near the **top-right**, long blank paper below; one delicate **orchid leaf** as a low counterweight at bottom-left. **Ink corridor width ≈ 0.20 page, margin ratio L:R $\approx 2:1$.** Cursive grass script with a few dry-brush catches. Two cinnabar seals: small **mid-right** and **bottom-left**. Cadence **punctuated** (short phrase, pause). No frame or Western layout.
- **S2B: Full Column Calligraphy (packing-led text):** Tall hanging scroll on cream washi with visible fibers; monochrome ink; faint deckled edges; subtle paper shadows. **One continuous column of cursive grass script** from top to near bottom; even spacing; occasional micro-splatter. **Ink corridor width $\approx 0.35\text{--}0.40$ page, margin ratio L:R $\approx 1:1$.** Title tag at top with a small cinnabar seal; second seal near the tail. Cadence **steady**. No frame or Western layout.
- **S3A: Waterfall Minimal (void-led image):** Tall hanging scroll on cream washi; monochrome ink; soft bleed. **Single vertical water stroke** descending into a blank basin; one small **reed tuft** as stopper on the lower right. **Ink corridor width ≈ 0.20 page, margin ratio L:R $\approx 1:2$.** Sparse rock texture dots, lots of untouched paper. Cinnabar seals: **mid-left** and **bottom-right**. Cadence **punctuated**. No frame.
- **S3B: Waterfall Constructed (packing-led image):** Tall hanging scroll on cream washi; monochrome ink; soft bleed. Layered **mountain waterfall** with a pine ledge; rock texture dots (cun), mist washes, multiple ink tones. **Ink corridor width $\approx 0.40\text{--}0.45$ page, margin ratio L:R $\approx 1:1$.** Two cinnabar seals **mid-left** and **bottom-right**. Cadence **steady**, dense mark energy. No frame.
- **S4A: Perched Bird & Empty Sky (void-led)**
Tall hanging scroll on cream washi; monochrome ink; faint deckled edges; paper shadows. A **single perched bird** on a thin twig entering from **lower-left**, pointing diagonally **into blank space**. **Ink corridor width ≈ 0.22 page, margin ratio L:R $\approx 2:1$.** One cinnabar seal **upper-right** as a distant counterweight; another **bottom-left**. Cadence **punctuated** (few strokes, long pause).
- **S4B: Flock & Reeds (packing-led):** Tall hanging scroll on cream washi; monochrome ink. **Cluster of small birds** above **reed thicket**, overlapping leaves and stems. **Ink corridor width ≈ 0.42 page, margin ratio L:R $\approx 1:1$.** Texture dots and layered reed fans. Seals **upper-left** and **lower-right**. Cadence **steady**, moderate density.
- **S5A: Scholar's Rock, Breathing Field (void-led);** Tall hanging scroll on cream washi; monochrome ink with soft washes. A **scholar's rock** silhouette on **lower-left**, rough dry-brush edge; wide breathing field to the right. **Ink corridor width ≈ 0.25 page, margin ratio L:R $\approx 2:1$.** A tiny inscription near upper-right; seals **bottom-left** and **mid-right**. Cadence **punctuated**.
- **S5B: Scholar's Rock with Moss Texture (packing-led);** Tall hanging scroll on cream washi; monochrome ink; rock rendered with **dense texture dots** and wet-over-dry layering; small **orchid clump** adjacent. **Ink corridor width ≈ 0.45 page, margin ratio L:R $\approx 1:1$.** Two seals **upper-left** and **lower-right**. Cadence **steady**, high packing.
- **S6A: Pine Cliff, Narrow Corridor (void-led);** Tall hanging scroll on cream washi; monochrome ink; **pine branch** enters from **upper-left**, needles pointing inward across empty paper; faint cliff edge implied. **Ink corridor width $\approx 0.22\text{--}0.25$ page, margin ratio L:R $\approx 2:1$.** Seals **mid-right** and **bottom-left**. Cadence **punctuated**; mark economy.
- **S6B: Pine Cliff, Constructive Stroke (packing-led);** Tall hanging scroll on cream washi; monochrome ink; **constructed cliff** with stacked planes, **pine platform**, textured rock faces. **Ink corridor width ≈ 0.45 page, margin ratio L:R $\approx 1:1$.** Repeated constructive strokes (cun) give dense surface. Seals **upper-left** and **bottom-right**. Cadence **steady**, high packing.
- **S7A: Tiger in bamboo, mist and diagonal light (packing-led image):** Tall hanging scroll on cream washi with visible fibers; **monochrome sumi-ink** wash, soft feathering/bleed, faint deckled edges, slight paper warp and shadows. A **crouching tiger** half-emerges from **bamboo stalks** at the left; stripes rendered with dry-brush texture and dot clusters; whiskers in quick flicks; diagonal mist/light bands drift across the blank

field. **Ink corridor width $\approx 0.38\text{--}0.42$ page, margin ratio L:R $\approx 1:1$** (bamboo + tiger occupy the central lane). Two small **cinnabar seals**: one **mid-left** near the bamboo joints, one **lower-right** as a counterweight. Stroke rhythm **steady** with denser mark energy around the head and paws; background washes light and layered. **No frame, no Western poster symmetry, no heavy color, no glossy gradients, no drop shadows, no modern fonts.**

- **S7B: Tiger in bamboo (void-led minimal):** Tall hanging scroll on cream **washi** with visible fibers; **monochrome sumi-ink** wash, soft feathering/bleed, faint deckled edges, slight paper warp and shadows. A **tiger's head and forepaws** emerge lightly from mist at lower left; only **two bamboo stalks** and a few leaves suggest the grove; wide open field to the right. Stripes in broken dry-brush; whiskers in a few quick flicks; pale ground wash disappears into paper. **Ink corridor width $\approx 0.22\text{--}0.26$ page, margin ratio L:R $\approx 2:1$** (very wide right margin). Two small **cinnabar seals**: **bottom-right** and **mid-left** near the bamboo as counterweights. Stroke rhythm **punctuated** (few accents, long pauses). **No frame, no Western poster symmetry, no heavy color, no glossy gradients, no drop shadows, no modern fonts.**
- **S8A: Wandering monk with calligraphy column (packing-led figure + text):** Tall hanging scroll on cream **washi** with visible fibers; **monochrome sumi-ink**; faint deckled edges; subtle paper shadows. A **wandering monk** strides from left to right in a sweeping robe, brush lines bold then tapering; boots dotted in dry-brush; the hem arcs forward. Above and to the right runs a **single column of cursive calligraphy** with a few watery ghost strokes behind it. **Ink corridor width $\approx 0.35\text{--}0.40$ page, margin ratio L:R $\approx 1:1$** (figure and text share one vertical lane). Two **cinnabar seals**: one **upper-right** beside the calligraphy; one **lower-left** near the robe's hem. Stroke rhythm **punctuated-then-steady** (gestural figure accents, even text cadence). **No frame, no Western layout, no typefaces, no color beyond seals, no vignette.**
- **S8B: Wandering monk with terse inscription (void-led):** Tall hanging scroll on cream **washi** with visible fibers; **monochrome sumi-ink**; faint deckled edges; subtle paper shadows. A **wandering monk** indicated with **economy of line**—a single sweeping contour for the robe, a few dry-brush marks for beard and boots—moving into the blank field. Only **two short lines of cursive calligraphy** near the **top-right**, ghosted by watery echoes; large open space below. **Ink corridor width $\approx 0.22\text{--}0.25$ page, margin ratio L:R $\approx 2:1$** (wider right). Two **cinnabar seals**: **mid-right** beside the inscription and **bottom-left** near the robe hem. Stroke rhythm **punctuated** (gesture, pause, gesture). **No frame, no Western layout, no typefaces, no color beyond seals, no vignette.**

Appendix F: Rolling Perspective Hypothesis / Follow Up

Handscrolls (horizontal) to test format-responsive corridor_90 (TEL); prompt interventions ('use full width') to test controllability.

Future work: The corridor width metric might be detecting whether the composition understands the scroll as a **temporal format and requires follow up**.

Defined: Rolling perspective (continuous perspective) does two things in eastern art.

- Treats the horizontal axis as the dimension of narrative unfolding. As the scroll opens right-to-left, the viewer's position moves through space and time. Exhibits variable viewpoints, narratives unfolding and multiple moments of time. Elements aren't seen simultaneously, they're revealed sequentially. Compositional relationships happen across that horizontal extent.
- Vertical however, this format is meant to be displayed on a wall and viewed in its entirety at once. The composition and perspective are designed to be appreciated as a whole rather than in a rolling sequence, still can exhibit variable viewpoints, multiple moments of time or a "scroll" down sense of perspectives in terms of overlapping planes.

If artists are using 76% of the page width (corridor_90 = 0.761) on vertical scrolls, they're distributing elements horizontally so those relationships can develop as the scroll unrolls. The vertical dimension is for local arrangement within each "moment," but the horizontal dimension carries the compositional logic.

The AI's 54% corridor suggests it's treating the scroll as a **static rectangle**, like a vertical Western painting. It centers a composition and adds margins. It's not thinking about horizontal distribution as temporal sequencing.

The void strategy difference reinforces this:

- **Artists:** Horizontal distribution (wide corridor) + vertical breathing (void through spacing). This creates pause and rhythm as you move through the scroll.

- **AI:** Vertical confinement (narrow corridor) + horizontal margins (void at edges). This creates a centered image with borders. It's spatial, not temporal.

The independence of corridor width and void for artists ($\rho \approx 0$) might reflect that these are different decisions:

- **Width** = how much horizontal extent does this compositional idea need
- **Void** = how much breathing room within that extent

The AI couples them weakly (narrow $\rightarrow$ less void) because it's thinking "content density" rather than "temporal pacing."

The caveat (which is why it requires a new dataset): This dataset is mostly hanging scrolls (vertical format), not handscrolls (horizontal format). Rolling perspective is most explicit in handscrolls where you literally unroll horizontally. Hanging scrolls are viewed all-at-once, but they often retain compositional conventions from handscroll tradition, vertical movement through landscape, with "near/far/middle" distributed top-to-bottom rather than temporally.

So the corridor_90 (TEL) metric might be detecting something about **distributional thinking** more generally, using the full spatial extent of the format as compositional space, whether that's temporal (handscroll) or vertical-depth (hanging scroll). The AI's narrow corridor suggests it's not thinking about format-specific spatial logic at all. It's just "put content in the middle." Testing this: If you run actual handscrolls (horizontal format), does artist corridor_90 go even higher? And does the AI stay narrow regardless of aspect ratio?

If hanging scrolls use the vertical axis for that shifting-perspective journey, then the corridor_90 (TEL) metric is detecting something different than I described.

What wide corridor (0.76) means for vertical scrolls:

Artists are distributing elements horizontally *to support* the vertical reading. As the viewer's eye travels up/down through the composition with shifting viewpoints, the horizontal placement creates:

- **Visual anchors** at different widths that guide the eye's vertical path
- **Counterpoint** between left and right elements that creates depth and spatial relationship
- **Breathing room** that allows the vertical journey to develop without lateral confinement

The horizontal dimension isn't carrying the narrative, it's providing the *spatial structure* within which the vertical narrative unfolds.

What narrow corridor (0.54) means:

The AI is making a centered vertical stack. It's not using horizontal distribution to support the vertical journey. Everything clusters in a middle column, so there's no lateral spatial complexity, just "content stripe with margins."

This means the AI doesn't understand that even in a vertically-read format, the horizontal dimension is compositionally active. It's treating width as "where to put the edges" rather than "how to structure the spatial field through which the viewer's perspective moves."

The void independence ($\rho \approx 0$ for artists) makes more sense now:

- **Corridor width** = how much horizontal extent to use for spatial structuring
- **Void** = how much breathing room in the vertical journey

These are independent decisions because they operate on different axes of the composition. The AI couples them because it's thinking "content blob size" rather than "vertical journey with horizontal spatial support."

So corridor_90 (TEL) is detecting whether the generator understands the format's dimensional logic: vertical scrolls need horizontal distribution to create the spatial field for perspective shifts.

Artists consistently produce wide corridors (0.76) while AI produces narrow ones (0.54), and for artists this is independent of void amount. We're *interpreting* this as "artists understand format-specific spatial logic" but the system itself just reports the numbers.

The correlation (or lack of it) between corridor width and void might be a useful proxy for compositional sophistication, but it's not directly measuring artistic intent or format understanding. It's finding a statistical pattern that seems to align with those concepts.

So: **The metrics detect a structural difference. This is only suggestively offering an explanation for why that difference matters.** corridor_90 (TEL) shows spatial distribution, not "format intelligence," but the pattern might correlate with it.

The question is whether the correlation holds across more data. Does corridor width reliably distinguish format-aware composition from generic centering? That's an empirical claim you'd need more samples to test. The delta is suggestive and worth pursuing. Here's what the data supports:

- **Strong observation:** Artists consistently use wider corridors (0.76) than AI (0.54). This is a 41% difference with clean separation between the groups. That's a real structural pattern.
- **Plausible hypothesis:** Wide corridors reflect format-aware spatial distribution (using horizontal extent to structure vertical composition). Narrow corridors reflect generic centering (content blob with margins).

What would validate this:

1. **Format variation test:** Run handscrolls (horizontal format). If artists go even wider in corridor_90 for horizontal formats, that supports the hypothesis that corridor width tracks format-specific spatial logic. If it stays the same or goes down, the hypothesis weakens.
2. **Controlled AI prompt test:** Explicitly prompt AI to "distribute elements across the horizontal extent" or "use full width of composition." If corridor_90 increases, the AI can do it but doesn't default to it. If it stays narrow, the model may have a structural limitation.
3. **Non-scroll baseline:** Run Western vertical paintings or other non-scroll formats. If Western paintings also show wide corridors (artists using horizontal space even in non-scroll formats), then corridor width might just be "spatial sophistication" generally, not scroll-specific. If Western paintings show narrower corridors than scrolls, that supports format-specificity.
4. **Within-artist variation:** If you have multiple works by the same artist in different formats, does their corridor_90 shift with format? If yes, it's format-responsive. If no, it might be personal style.

What could weaken the hypothesis:

If wider corridors just correlate with "more elements in the composition" rather than format understanding, then it's detecting complexity, not sophistication. You'd want to check: Do pieces with fewer elements (minimal compositions) still show the corridor difference? If minimal artist scrolls are also at 0.76 while minimal AI scrolls are at 0.54, that's stronger evidence for format logic rather than just "more stuff."

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